

DEM modeling of ball mills with experimental validation: influence of contact parameters on charge motion and power draw

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Abstract Discrete element method simulations of a 1:5-scale laboratory ball mill are presented in this paper to study the influence of the contact parameters on the charge motion and the power draw. The *position density limit* is introduced as an efficient mathematical tool to describe and to compare the macroscopic charge motion in different scenarios, i.e. with different values of the contact parameters. While the charge motion and the power draw are relatively insensitive to the stiffness and the damping coefficient of the linear spring-slider-damper contact law, the coefficient of friction has a strong influence since it controls the sliding propensity of the charge. Based on the experimental calibration and validation by charge motion photographs and power draw measurements, the descriptive and predictive capabilities of the position density limit and the discrete element method are demonstrated, i.e. the real position of the charge is precisely delimited by the respective position density limit and the power draw can be predicted with an accuracy of about 5 %.

Keywords Ball mill · Discrete element method · Contact parameters · Experimental validation

1 Introduction

Comminution, i.e. particle size reduction, is an essential component of mineral processing and cement production. *Grinding* is the final stage of comminution consisting in breaking already small particles into much smaller ones (several mm to several μm) in order to enhance their chemical and physical properties. A major category of grinding devices are *tumbling mills*, i.e. rotating cylindrical drums filled with the charge, which consists of the feed material and the grinding media. In particular, *ball mills* are tumbling mills whose grinding media are metal balls. A profiled and wear-resistant *liner* is installed in the mill to transfer energy to the grinding charge more efficiently and to protect the mill shell [47].

Grinding in ball mills is the most *energy-intensive* comminution operation; for instance, about 110 kWh of electrical energy are consumed per ton of cement produced. Around 70 % of this energy is needed to grind the raw material and the clinker, while only 1–5 % of the supplied energy is needed for the explicit particle breakage [3, 20], i.e. the remaining energy is wasted. For this reason, even the slightest improvement in terms of energy efficiency can lead to a considerable economic impact.

Optimization of milling processes by the classical way, i.e. *experimental testing*, is difficult and costly due to the harsh milling environment and the size of these mills (diameter of about 4 m, 5–10 MW power draw, 300 ton/h throughput). In addition to the rising computational performance of modern desktop computers and the limitations of semi-analytical models [34], this is why the *discrete element method* (DEM) [9] has become a popular alternative way to solve this problem.

Two- and three-dimensional DEM simulations of ball mills were first described in [27] and [8], respectively.

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Detailed reviews about this subject can be found in [25,26] and more recently in [46].

Despite the study of ball mills by the discrete element method for more than two decades, no general consensus about the *soft contact law* (Sect. 2.2.2) and its *material parameters* seems to exist. More specifically, the *linear spring-slider-damper contact law* [45] as well as the *non linear Hertz–Mindlin law* [15,23,24] were extensively used in the context of DEM simulations of tumbling mills, e.g. [3,12,18,27] and [21,22,32,37], respectively. While the first model is mainly elected by researchers because of its simplicity, which reduces the computation time, the second model generally appeals due to its better agreement with the idealized contact mechanics [41], i.e. the analytic solution of the collision problem of two elastic spheres [15,23,24].

Moreover, the values of the contact parameters also feature a large scattering in the literature even though the ultimate objective is to study mills containing only metal balls and rocks. *Stiffness coefficients* commonly take values between 10^4 and 10^8 N/m depending on experimental measurements [27] and the maximum acceptable overlap of colliding bodies in order to reduce the computation time [7] (explicit time integration, Sect. 2.2.3). *Coefficients of restitution* are assumed to have a value between 0.3 and 0.8 [3,6,10,27,35]. Furthermore, different values are sometimes used for ball-ball and ball-liner collisions [19,42]. And finally, the *coefficient of friction* also spans a wide range in the literature: 0.1 in [27], 0.14 in [13], 0.2 in [17], 0.4 in [11], 0.5 in [6,35], 0.6 in [10] and 0.75 in [3], ...

The previous reasons make the question rise whether this *complexity by diversification* is necessary to predict the charge motion and the power draw of ball mills. In this work, it is therefore shown that the simple *linear spring-slider-damper soft contact law* is sufficient to model the previous characteristics. Moreover, the *influence of the contact parameters* on the charge motion and the power draw is studied by introducing the *position density limit*, which is a mathematical tool enabling the objective comparison of the macroscopic charge motion in different scenarios. In order to prove the predictive capability of the model, the results of the simulation are calibrated and validated on the basis of charge motion photographs and power draw measurements taken from a *1:5-scale model of a ball mill* kindly provided by the company *Magotteaux International s.a.* Never before, validation data in a mill of that size has been published. Validation studies with smaller mills can be found in [7,42,43].

First, the general *methods*, i.e. the experimental setup, the discrete element method and the position density limit, are described in Sect. 2. Afterwards, the *results* are presented and discussed by illustrating the influence of the contact parameters on the charge motion and the power draw as well as by comparing the numerical results with the experimental measurements.

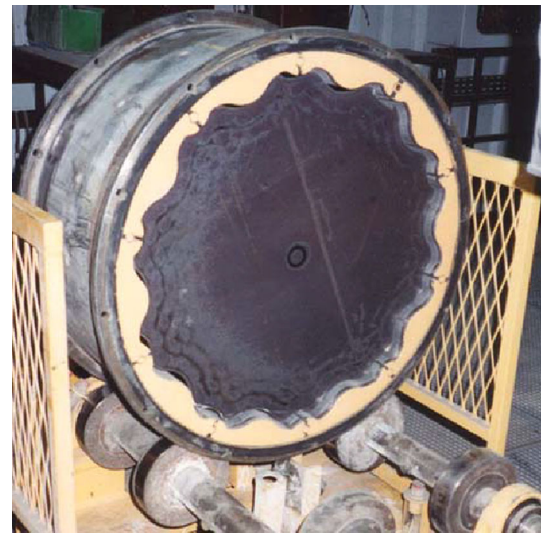


Fig. 1 Laboratory mill supported on driving shafts with the liner 2 (Fig. 3) and closed by an end wall of transparent tempered glass. Courtesy of Magotteaux International s.a. (Color figure online)

2 Methods

In this section, the *experimental setup* of the laboratory mill is described before the numerical methods, i.e. the *discrete element method* and the *position density limit*, in order to simplify the explanations by applying directly the general principles to the particular application.

2.1 Experimental setup

The experimental setup consists of a 1:5-scale model of a ball mill, more precisely, the first chamber of a cement grinding tube mill (Fig. 1). Its shell diameter and axial length are respectively equal to 800 and 400 mm. The mill is closed by flat end walls and a window of transparent tempered glass in order to take pictures of the charge motion. A torquemeter between the electric motor and the mill shell was used to measure the average power draw in the quasi-steady state, i.e. the state in which the macroscopic characteristics, like the center of gravity of the balls, do not change significantly anymore when the mill rotates with a constant speed. Two different geometries of the shell liner were studied: the liner 1 (Fig. 2) and the liner 2 (Fig. 3).

Concerning the operating conditions, the mill is filled with almost spherical chromium-alloyed steel balls having a diameter of about 15 mm. Their material and bulk density are respectively equal to 7640 and 4580 kg/m³. The mill does not contain any feed material, like crushed clinker nodules, since it is assumed that the charge motion is not significantly altered by the presence of this feed material. In fact, the clinker is substantially lighter than the chromium-alloyed steel, i.e. its density is equal to about 1400 kg/m³,

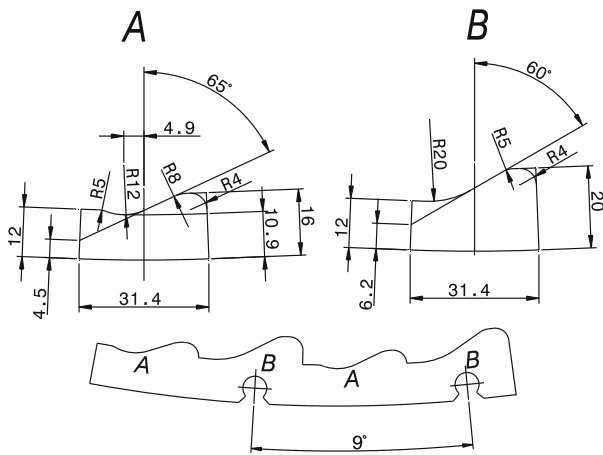


Fig. 2 Geometry of the liner 1 in (mm) (individual plate and stacking order). The profile moves from the *right* to the *left*, i.e. the mill rotates clockwise with respect to this representation

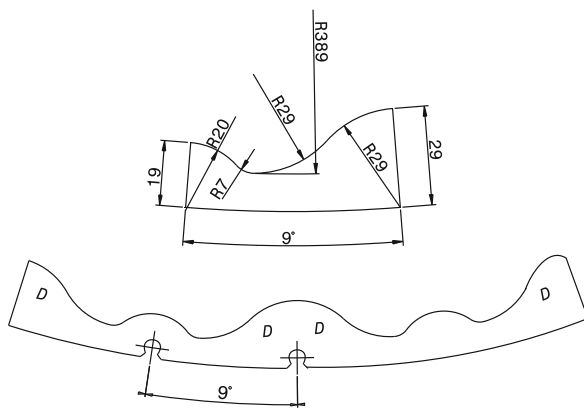


Fig. 3 Geometry of the liner 2 in (mm) (individual plate and stacking order). The profile moves from the *right* to the *left*, i.e. the mill rotates clockwise with respect to this representation

its size is also decisively smaller than the balls, i.e. several millimeters to several micrometers, and its volume in the mill is usually equal to the volume of the interstices between the balls. The *filling ratio* f , i.e. the bulk volume percentage with respect to the total volume of the mill, is calculated on the basis of the mean liner diameter, which is equal to 772 and 763.3 mm for the liner 1 and the liner 2, respectively. The rotation speed Ω will be deduced from the defined *percentage of critical speed* p_{cs} , i.e. the percentage with respect to the rotation speed for which its charge would be centrifuged, based on the mean liner diameter $\bar{D}$. More precisely, a ball is centrifuged when the centrifugal force acting on it becomes equal to its weight, i.e. $m \Omega_c^2 \bar{D}/2 = m g$, where m , Ω_c and g are respectively the mass of a ball, the critical rotation speed and the gravitational acceleration. The previous formula assumes that the ball diameter is significantly smaller than the mean liner diameter.

Table 1 Liner geometry, filling ratio f , percentage of critical speed p_{cs} , number of balls n and rotation speed Ω of the different calibration (case 1) and validation cases (cases 2, 3 and 4)

Case	Liner	f (%)	p_{cs} (%)	n (—)	Ω (rpm)
1	1	30	70	19,054	33.70
2	1	30	80	19,054	38.51
3	1	40	80	25,406	38.51
4	2	30	70	18,642	33.88

For the cases described in Table 1, pictures of the charge motion were taken and the power draw was measured during the quasi-steady state (Sect. 3.2).

2.2 Discrete element method

The discrete element method is a numerical Lagrangian meshfree method used to simulate mainly the kinematic and dynamic behavior of particulate systems, like ball mills. It is based on three fundamental components: *bodies*, *interactions* and the *time integration* [1, 9, 29, 33, 41]. The DEM simulations in this paper were carried out in 3D because of the limited shear strength of the particle structure and the neglected influence of the end walls in 2D [7]. At the end of this section, the output data of the simulations will also be briefly described.

2.2.1 Bodies

To simulate the charge motion in the ball mill, three different bodies were used: spherical elements representing the balls, triangulated surfaces representing the shell liner and two planes representing the flat axial end walls. While the triangular facets and the lateral planes are rotating with a constant angular velocity around the rotation axis of the mill, the balls are floating and rotating in space according to Newton's second law and interacting according to the contact law described in Sect. 2.2.2.

Initially, the mill is filled numerically by adding the precise number of balls to the lower horizontal cylinder segment. In consequence, the balls will first fall down during a fraction of a second under the influence of gravity before colliding with the liner. Due to its rotation, the liner will then lift up the charge before it cascades and cataracts down to the toe (Fig. 4). Finally, after about 2.5 s, the motion of the balls reaches the quasi-steady state for the four scenarios presented in Table 1. The criterion, which was chosen to determine when this quasi-steady state starts, is the time, when the center of gravity of all the balls does not evolve significantly anymore.

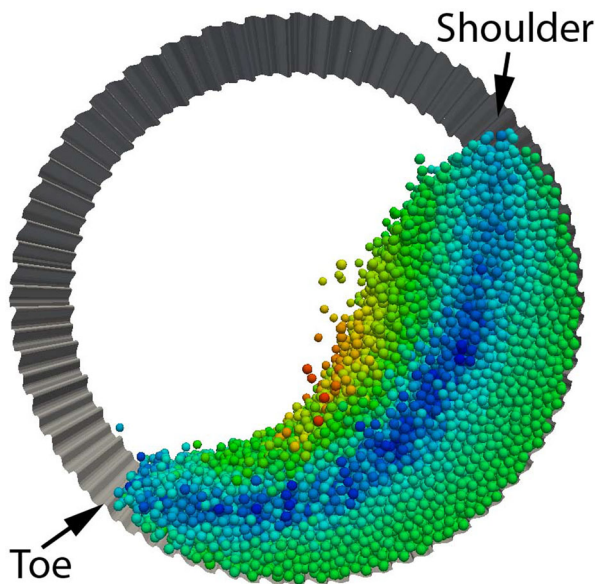


Fig. 4 Example of an isometric projection of the charge motion inside of the laboratory mill (case 1 of Table 1) in ParaView. The color code indicates the magnitude of the translational velocity: 0 m/s (blue) to 3 m/s (red). This picture shows the characteristic regions of the charge motion, i.e. the toe and the shoulder, as well as the stochastic variability of the ball flow pattern. (Color figure online)

2.2.2 Interactions

In order to prevent the balls, the facets and the lateral planes from overlapping completely with other balls when they collide, their contacts are detected and the collision forces and torques are determined.

The simulations in this paper were performed using *YADE* [40], an open-source DEM solver, due to its good performance, its versatility and its PYTHON interface. The standard *collision detection* algorithm in *YADE* is a sweep and prune insertion sort algorithm parallelized by an *OpenMP* shared memory multiprocessing approach. Once, this fast collision algorithm has detected potential contacts, an exact collision detection then determines the precise overlap of the colliding bodies, which is a consequence of the soft contact law.

In our simulations, this *soft contact law*, i.e. the colliding bodies are allowed to slightly overlap, is the linear spring-slider-damper contact law (Fig. 5). This model was chosen because of its simplicity and ability to simulate the energy transfer between bodies (springs), the sliding between particles (slider), the dissipation of energy during collisions (dampers) and the rotation of particles (tangential component). Even though the Hertz–Mindlin contact law approximates better the idealized contact mechanics, it was not chosen because its higher complexity was not necessary to predict the charge motion and the power draw (Sect. 3.2).

The linear spring-slider-damper contact law can be described mathematically by the following simplified formulas

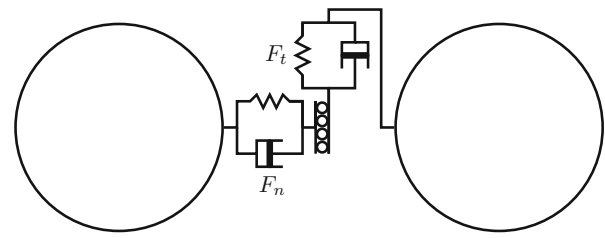


Fig. 5 Schematic representation of the linear spring-slider-damper contact law between two colliding bodies with its normal and tangential components F_n and F_t , i.e. a spring in parallel with a damper for each component, connected by a sliding device

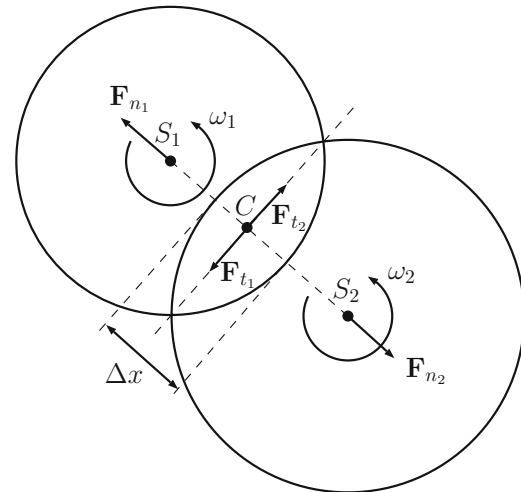


Fig. 6 Geometric representation of two colliding spheres S_1 and S_2 rotating as indicated by ω_1 and ω_2 . The resulting normal and tangential forces are $\mathbf{F}_{n1}$ and $\mathbf{F}_{n2}$ as well as $\mathbf{F}_{t1}$ and $\mathbf{F}_{t2}$, respectively applied to the center of S_1 and S_2 , and the contact point C of each sphere

for the normal and the tangential collision forces F_n ($F_n > 0$, if repelling) and F_t :

$$F_n = k_n \Delta x + c_n v_n \quad (1)$$

$$F_t = \min \left(\mu F_n, k_t \int v_t dt + c_t v_t \right) \quad (2)$$

where Δx , v_n and v_t are the overlap ($\Delta x > 0$, if overlapping, see Fig. 6), the normal relative velocity ($v_n > 0$, if approaching) and the tangential relative velocity of two colliding bodies. The directions of the vector forces created during a collision are given in Fig. 6 for two approaching spheres such that $|F_n| = |\mathbf{F}_{n1}| = |\mathbf{F}_{n2}|$ and $|F_t| = |\mathbf{F}_{t1}| = |\mathbf{F}_{t2}|$.

Moreover, the torque $\mathbf{T}$ created by the tangential contact force $\mathbf{F}_t$ puts the balls into rotation:

$$\mathbf{T} = \mathbf{SC} \times \mathbf{F}_t \quad (3)$$

where $\mathbf{SC}$ is the vector pointing from the center of the sphere to the contact point as shown in Fig. 6. The detailed mathe-

mathematical description of the linear spring-slider-damper contact law can be found in [2,31,44,45].

Despite the apparent simplicity of the model, it already contains five material parameters: the normal stiffness k_n , the tangential stiffness k_t , the normal damping coefficient c_n , the tangential damping coefficient c_t and the Coulomb friction coefficient μ . Notice that the normal damping coefficient c_n is commonly written in terms of the *coefficient of restitution* ε_n , if two spheres with their individual masses m_1 and m_2 , collide [2,3]:

$$c_n = -2 \ln(\varepsilon_n) \sqrt{\frac{m_{\text{eff}} k_n}{\pi^2 + \ln^2(\varepsilon_n)}} \quad \text{with} \quad m_{\text{eff}} = \frac{m_1 m_2}{m_1 + m_2} \quad (4)$$

The different terms in Eqs. (1) and (2) can be explained in more detail as follows. The first term on the RHS of Eq. (1) quantifies the force generated by the normal linear spring while the second term represents the force generated by the normal linear damper. In contrast to the normal spring, the compression of the tangential spring can not be directly deduced from the geometrical configuration. Hence, the tangential velocity is integrated and multiplied by the corresponding stiffness to compute the force generated by this spring in Eq. (2). The force created by the tangential damper is computed in a similar way than the force of the normal damper. Finally, the minimum function in Eq. (2) selects the Coulomb friction limit, i.e. μF_n , if the tangential force created by the spring and damper exceeds this limit.

Two important remarks have to be added to the previous explanation since some features do not appear explicitly in Eqs. (1) and (2). First, the normal contact force is set equal to zero, when the right hand side of Eq. (1) becomes negative; attractive forces would not be physical during the contact of metal balls. Strictly speaking, Eq. (4) is therefore not valid anymore because it is based on Eq. (1) without the sign condition concerning the normal force [39]. Since this problem is, however, only a matter of definition of the coefficient of restitution, Eq. (4) will still be used to determine the damping coefficient in this paper, like in [3]. Second, the time integral in Eq. (2), which is related to the tangential spring, is not a simple integral, since the spring stores and loses energy due to the relative motion between the elements. In consequence, the relative tangential velocity can not simply be integrated in global coordinates in order to determine the strain of the spring. This problem is solved by rotating the spring force vector simultaneously with the colliding elements [40].

In addition to the contact interactions between bodies, the balls are also subjected to gravity. The resulting force is therefore simply added to the contact forces. Once all the forces are applied to a ball, its acceleration, velocity and position at the following time step can be determined by Newton's sec-

ond law and the numerical time integration scheme, which is presented in the next section.

2.2.3 Time integration

The time integration scheme implemented in *YADE* is the conditionally stable second order accurate explicit *leapfrog algorithm*, which can be written as follows for the translation and rotation of the balls:

$$\dot{\mathbf{x}}(t + \Delta t/2) = \dot{\mathbf{x}}(t - \Delta t/2) + \mathbf{F}(t)/m \Delta t \quad (5)$$

$$\mathbf{x}(t + \Delta t) = \mathbf{x}(t) + \dot{\mathbf{x}}(t + \Delta t/2) \Delta t \quad (6)$$

$$\boldsymbol{\omega}(t + \Delta t/2) = \boldsymbol{\omega}(t - \Delta t/2) + \mathbf{T}(t)/I \Delta t \quad (7)$$

where $\mathbf{x}$, t , Δt , $\mathbf{F} = \mathbf{F}_n + \mathbf{F}_t$, m , $\boldsymbol{\omega}$, $\mathbf{T}$ and I are the position vector of a ball, the current time, the time step, the force acting on a ball, its mass, the angular velocity vector, the torque acting on a ball and its moment of inertia, respectively.

To ensure the stability of the algorithm, it is necessary that the time step is smaller than the *critical time* step Δt_c , e.g. $\Delta t_c = 2/\omega_{\text{max}}$ for an undamped system, where ω_{max} is the highest frequency in the system [30]. Furthermore, this time step can also be determined in order to limit the creation of numerical energy, i.e. to recover the coefficient of restitution. More precisely, the computed coefficient of restitution of a collision between two bodies in the DEM simulation converges to its analytical counterpart when the time step decreases. In other words, less artificial energy is created when the time step decreases. By limiting the creation of artificial energy in normal collisions to 1 % with respect to the total energy in the system, this criterion is more restrictive than the stability requirement. Due to the lengthy derivation of the criterion, the time steps used in the following simulations will only be mentioned here (Sect. 3.1); their detailed calculation can be found in [2]. In simple words, this criterion is equivalent to choosing 20–70 time steps per collision depending on the coefficient of restitution; more time steps per collision are usually required when the coefficient of restitution is important due to the higher propagation of artificial energy in this case.

2.2.4 Output data

In increasing order of measuring difficulty, the most used validation parameters of the discrete element method in the field of tumbling mill simulations are the power draw [5], the charge motion [7,14,36] and the forces on the liner [28].

Since our experimental setup renders only the kinematic and energetic validation possible, the output data of the simulations will be the position of the balls and the power draw of the mill. In the quasi-steady state, this power draw can be estimated in three different ways: by computing explicitly the

torque exerted by the balls on the liner and multiplying this torque by the rotation speed, by calculating the total power dissipation due to collisions or by determining the average torque exerted by the balls on the liner based on the center of gravity of the balls and multiplying it by the rotation speed. On average, these values were equal, i.e. their relative difference was smaller than 1 %, in our simulations during the quasi-steady state. The first definition was chosen to estimate the power draw since it is equal to the theoretical power draw in every scenario, i.e. also outside of the quasi-steady state.

2.3 Position density limit

Based on the previous explanations, it seems to be possible to predict the charge motion and the power draw of the ball mill by the discrete element method once the values of the contact parameters and the time step have been determined (Sect. 3.1). In contrast to the power draw, it is difficult to characterize the global motion of the balls during the quasi-steady state as shown in Fig. 4. Usually, the motion is therefore only described by the toe angle, i.e. the position down to which the balls cascade, and the shoulder angle, which is the highest point reached by the charge [7].

Due to the stochastic variability of the charge motion at the scale of the balls, comparing this motion for different operating conditions, different liners or different material parameters in the simulation (Sect. 3.1) is difficult, imprecise and subjective. For this reason, the position density limit will be introduced hereinafter.

The *position density limit* is a mathematical tool enabling the objective comparison of the macroscopic charge motion in different scenarios. More precisely, it is a specific contour curve of the position density plot, which quantifies the number of occurrences of balls at a particular position in the mill [16].

To determine the position density limit of a particular milling scenario, the space taken up by the mill in the xz -plane (y is aligned with the rotation axis) is discretized by a rectangular plane grid (Fig. 7). For each cell of this grid, the number of balls having the projection of their center in this specific cell is accumulated for all time steps, once the quasi-steady state has been reached. After dividing this number by the total number of time steps (in the quasi-steady state), we obtain the average number of balls per cell. These numbers are represented in Fig. 8, which represents the *position density plot* of a specific configuration.

Instead of accumulating the ball occurrences for each time step Δt , it is even more efficient to record these occurrences at a smaller sampling frequency, i.e. every $\Delta t_{\text{rec}} > \Delta t$, since the displacement of a ball during one time step is relatively insignificant. Using more cells along each direction of the grid obviously refines the position density plot but at the same time the sampling duration, i.e. the time during which

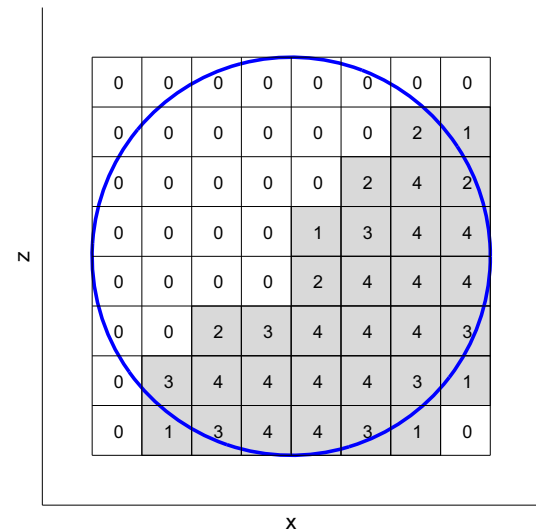


Fig. 7 Example of a 8×8 —binning grid, which is used to accumulate the ball occurrences in the mill. This mill is represented by the circle. The numbers represent the hypothetical average number of ball occurrences in each cell

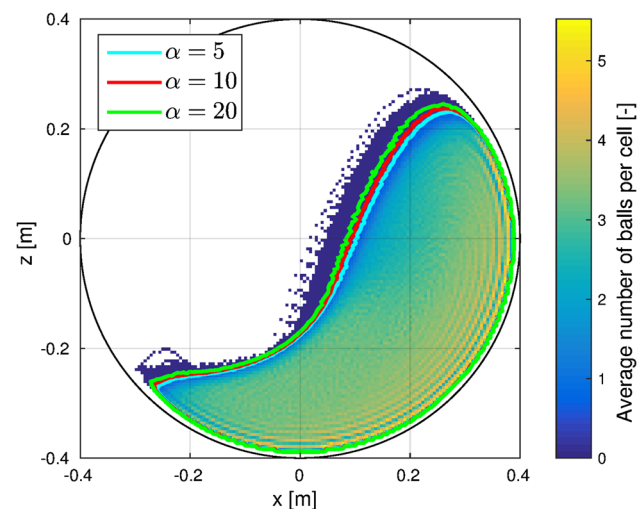


Fig. 8 Example of a position density plot of the charge motion in the quasi-steady state obtained with a 150×150 —grid (case 1 of Table 1). The respective position density limits for different limiting factors α are also represented. (Color figure online)

the positions are sampled every Δt_{rec} , has to be increased in order to obtain sufficiently representative data. One may notice that the position density plot can be interpreted as the estimated probability of ball occurrences at a particular position of the mill.

Finally, the *position density limit* is the contour curve, which separates the main charge zone from the region where the balls are less likely to occur. The contour curve can be constructed by the *marching squares algorithm* [38], once its level has been fixed. This level was chosen equal to the average of the average number of balls per cell over all the

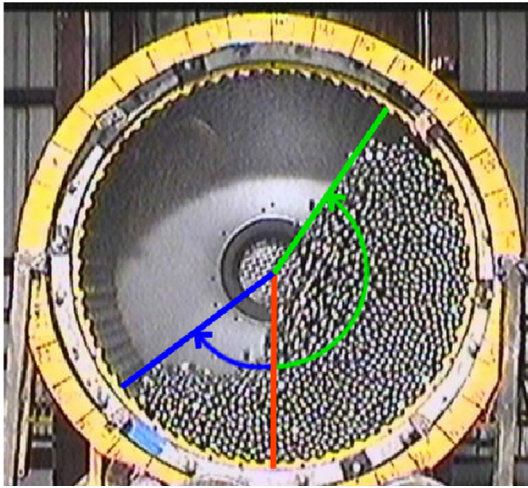


Fig. 9 Illustration of the shoulder and toe angle measurement based on a photograph of the charge motion in the laboratory mill. (Color figure online)

non-zero occurrence cells divided by $\alpha = 10$. In the case of the example represented in Fig. 7, this level is actually equal to $96/32/10 = 0.3$, since 96 is the total number of balls (sum of all the numbers in the cells) and 32 is the number of cells with a number of ball occurrences different from 0.

It is important to calculate the average with respect to the cells in which occurrences of balls have actually been accumulated in order to take into account the degree of dispersion of the charge cloud. Indeed, if the average would be calculated with respect to all the cells, i.e. also the empty cells, it would be constant for a given grid since the total number of balls does not change. At a higher rotation speed, the charge cloud is, however, more expanded than at a lower one and therefore the average over all the cells would be too restrictive. Despite this improvement of the level criterion, it still remains partially subjective because of the factor $\alpha = 10$. The choice of this factor is based on the classical measurement strategy of the shoulder and toe angles by laboratory experiments. In fact, these angles are measured on photographs of the charge motion as shown in Fig. 9. Due to the stochastic motion of the balls, the measurement of these angles is always subjective since one has to visually separate a zone where balls are likely to occur from a zone where the balls are less likely to occur. A reasonable assumption is that the experimenter separates the main charge region on photographs from the region where the balls are on average about one order of magnitude less likely to occur, i.e. 10 times less likely. Moreover, Fig. 8 shows the relatively low sensitivity of the position density limit to the limiting factor α provided that a sufficient decrease in the position density is imposed to separate the regions.

Unlike the position density plot, position density limits have the benefit of simplifying the charge motion description tremendously. Toe and shoulder angles can be measured eas-

Table 2 Default values of the contact parameters

Contact parameters	Variable	Value
Normal stiffness	k_n	10^5 N/m
Tangential stiffness	k_t	k_n
Normal coefficient of restitution	ε_n	0.3
Tangential damping coefficient	c_t	c_n
Friction coefficient	μ	0.75

The normal damping coefficient c_n is deduced from the normal coefficient of restitution ε_n by the Eq. (4)

ily and relatively objectively. And what is even more important is the possibility of superimposing different position density limits obtained with different operating conditions or different contact parameters in order to study their influence on the global charge motion. This latter case is studied in the following section.

3 Results and discussion

Previously, the experimental setup, the discrete element method and the position density limit were described. In this section, these methods are used to study the influence of the contact parameters on the charge motion and the power draw. Moreover, the best fitting parameter set will be determined by experimental calibration and validation to cross-check whether the position density limit describes accurately the charge motion.

3.1 Influence of the contact parameters

In order to determine the influence of the contact parameters on the results of the simulation, the change of the position density limit and the power draw will be studied as a function of the contact parameters in a specific scenario. This scenario is case 1 of Table 1. The values of the parameters in this scenario are standard values used in the cement grinding industry. Unless otherwise stated, we suppose that the contact parameters take the values given in Table 2; these choices will be explained hereinafter. Notice also that the same values were used for ball-ball and ball-liner collisions.

The *time step* was chosen equal to $\Delta t = 10^{-5}, 2.5 \cdot 10^{-6}$ and 10^{-6} seconds respectively for $k_n = 10^5, 10^6$ and 10^7 N/m, in order to limit the percentage of artificial numerical power with respect to the total power to 1 %, as explained in Sect. 2.2.3. Strictly speaking, more than three different values of the time step should be used since this time step also depends on the damping coefficient, i.e. not only on the stiffness. In order to simplify the explanations, the smallest time step for a given stiffness was, however, chosen.

The position density limits and the average power draws will be calculated between the moments $t = 2.5$ s and $t = 5$ s, since the system reaches the quasi-steady state after 2.5 s according to the position of the global center of gravity; 2.5 additional seconds are necessary to obtain sufficiently representative averages. These averages are based on the data collected every $\Delta t_{\text{rec}} = 0.01$ s, which is the recording period. The corresponding position density limits were determined on a 150×150 -grid since it leads to a sufficiently detailed (more cells) and statistically meaningful (less cells) representation of the position density.

In the following sections, the importance of the *tangential component* of the contact law and the influence of the different contact mechanisms, i.e. *stiffness*, *damping* and *friction*, will be studied.

3.1.1 Influence of the tangential component

Using different values for the normal and tangential stiffness and damping coefficients, i.e. $k_n \neq k_t$ and $c_n \neq c_t$, requires a very detailed validation study due to the important number of a priori independent parameters: k_n , k_t , c_n , c_t and μ . Therefore, it seems literally impossible to calibrate these parameters independently from each other. Nevertheless, it was attempted to understand their influence on the results by adjusting the relative importance of the tangential spring and damper with respect to their normal counterparts.

According to the idealized contact mechanics [41], the normal and tangential stiffness have to satisfy the following condition: $2/3 \leq k_t/k_n \leq 1$. Nonetheless, we also considered the case $k_t = 4/3 k_n$ to test whether a greater tangential stiffness than the normal stiffness would have any influence on the results. Moreover, similar definitions were used to define the tangential damping coefficient c_t based on the normal damping coefficient c_n , i.e. that the position density limits and the power draws for the nine possible combinations of $k_t = 2/3 k_n$, k_n and $4/3 k_n$ as well as $c_t = 2/3 c_n$, c_n and $4/3 c_n$ were determined.

All the resulting *position density limits* were not represented since they are completely superimposed. Hence, the change in the importance of the tangential stiffness and damping with respect to their normal counterparts has no significant influence on the global charge motion. Nevertheless it might have some influence on the spin of the particles. Only the average contour of the ball charge can, however, be experimentally validated by photographs, i.e. not the spin. Concerning the *power draw*, no significant change can be observed since the relative error between the maximal and minimal power is smaller than 1 %.

Consequently, the tangential component of the contact model has no significant influence on the charge motion and the power draw within the considered parameter space. It is, however, necessary to be incorporated in the model to

dissipate energy along the tangential direction and to reach the Coulomb friction limit as will be shown in Sect. 3.1.4. The most straightforward choice is therefore to assume that $k_t = k_n$ and $c_t = c_n$. Hence, the contact law only depends on the three remaining parameters: the normal stiffness k_n , the normal damping coefficient c_n and the coefficient of friction μ , which will be analyzed in the following sections.

3.1.2 Influence of the stiffness

A common technique to choose the value of the normal stiffness k_n is to limit the average overlap between two colliding spheres to a value between 0.1 and 1 % with respect to the diameter of the ball [4]. There is, however, no perfect value of overlap; maybe 10 % would still lead to satisfying results even though the geometrical interaction seems to be significantly altered in comparison with the reality where steel balls never overlap that much. In fact, the maximum overlap can be written as a function of the relative impact velocity v_r and the contact parameters by solving the respective analytic problem based on Eq. (1) [2]:

$$\Delta x_{\max} = \frac{v_r}{\omega_d} e^{-\zeta \omega_0 t^*} \sin(\omega_d t^*)$$

$$\text{with } t^* = \frac{1}{\omega_d} \arctan\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right) \quad (8)$$

where the natural frequency ω_0 , the damping ratio ζ and the damped frequency ω_d are defined by:

$$\omega_0 = \sqrt{\frac{k_n}{m_{\text{eff}}}} \quad \zeta = \frac{c_n}{2\sqrt{k_n m_{\text{eff}}}} \quad \omega_d = \omega_0 \sqrt{1-\zeta^2} \quad (9)$$

Since the effective mass m_{eff} and the normal damping coefficient c_n were defined in the Eq. (4), it is possible to compute the maximum overlap percentage of two colliding balls based on the normal stiffness k_n and the normal coefficient of restitution ε_n , provided that the maximum relative impact velocity v_r is known. This velocity can be assumed to be equal to $\sqrt{2 \times 9.81 \times 0.8} \approx 4$ m/s, which corresponds to the terminal velocity when a ball falls from the top of the mill (diameter of 0.8 m) to its lowest position starting from rest. The respective maximum percentage of overlap is represented in Fig. 10.

Furthermore, the desire to decrease the overlap is mitigated by the necessary reduction of the time step when the stiffness increases. Hence, the normal stiffness should be the greatest value, which induces no significant errors in the DEM solution, in order to minimize the computation time.

Consequently, it seems reasonable to choose the lower limit of the normal stiffness equal to 10^5 N/m because the percentage of overlap is then roughly smaller than 8 % for each

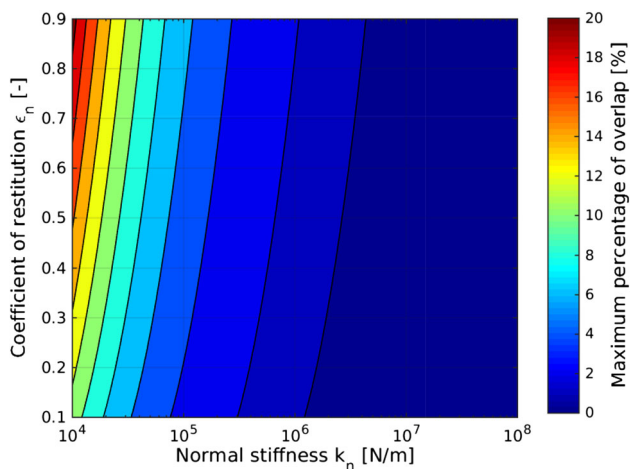


Fig. 10 Maximum percentage of overlap with respect to the diameter (of one sphere) during the normal collision of two (freely moving) spheres obtained with the linear spring-slider-damper contact law. The spheres have a 15 mm diameter and a density equal to 7640 kg/m³. The relative normal impact velocity is equal to 4 m/s. The frontiers in the contour diagrams are associated to the following percentages: 1, 2, 4, 6, 8, 10, 12, 14, 16, 18 and 20 %. (Color figure online)

value of the coefficient of normal restitution. If the sphere collides, however, with a facet, this percentage of overlap is even higher, i.e. around 10 % instead of 8 %, since $m_{\text{eff}} = m$ in this case, where m is the mass of the sphere. It should be noticed that the previous reasoning is based on the maximum overlap, i.e. on the maximum velocity equal to 4 m/s. On average, these velocities are, however, far smaller than 4 m/s.

Concerning the upper limit of the normal stiffness, a value greater than 10^7 N/m seems to lead to no significant reduction of the overlap, which is anyways smaller than 1 % for this value. Hence, the charge motion and the power draw will be determined for $k_n = 10^5$, 10^6 or 10^7 N/m; the values of the other parameters are given in Table 2.

Figure 11 shows the influence of the normal stiffness k_n on the global charge motion for identical values of the coefficient of normal restitution $\varepsilon_n = 0.3$ and the friction coefficient $\mu = 0.75$. Similar plots were, however, also created for all the combinations of $\varepsilon_n = 0.3$, 0.6 and 0.9 as well as $\mu = 0.25$, 0.5 and 0.75. These plots were not represented here since the respective conclusions are the same than for Fig. 11. Hence, the global charge motion in the quasi-steady state is almost independent of the normal and tangential stiffness ($k_t = k_n$, see Table 2); only the toe region slightly decreases when the stiffness increases. One can assume that the ball structure is slightly weaker, when the stiffness decreases and therefore the balls cascade preferentially to the toe.

As a consequence, the power draw increases for greater values of the stiffness (and unchanged values of ε_n and μ). In this particular case, the average power draw in the quasi-steady state is equal to 1391, 1414 and 1410 kW, for

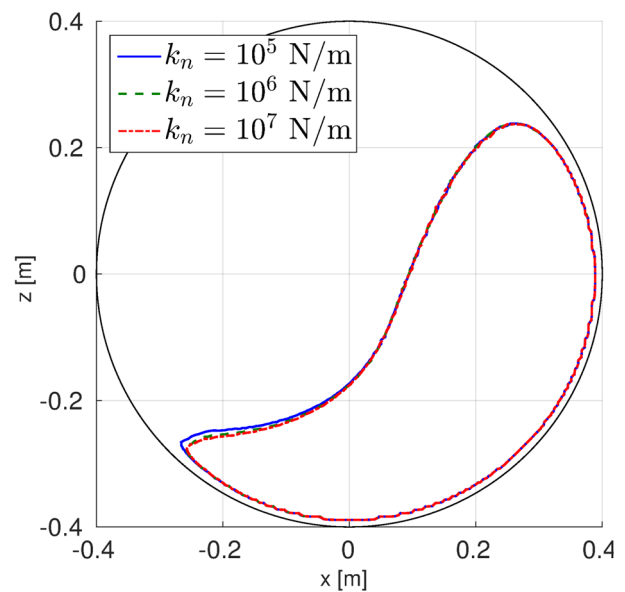


Fig. 11 Influence of the normal stiffness k_n on the global charge motion. This figure shows the outline of the mill shell and the position density limits for different values of the stiffness as well as for unchanged values of the coefficient of normal restitution $\varepsilon_n = 0.3$ and the coefficient of friction $\mu = 0.75$. (Color figure online)

$k_n = 10^5$, 10^6 and 10^7 N/m, respectively. The last value is only greater than the second value because of the limited sampling duration, which still contains a small transitory variation. In general, the influence of the stiffness on the global charge motion and the power draw is, however, negligible for the given parameter set, i.e. if the maximum percentage of overlap remains smaller than 8 % (resp. 10 %) for a free ball/ball (resp. ball/facet) collision.

The cases with even higher percentages of overlap were not studied since 8 % (resp. 10 %) is already a significant maximum percentage of overlap.

3.1.3 Influence of the damping

In the same way than for the stiffness, three different values were chosen for the coefficient of normal restitution ε_n , which is used to determine the normal damping coefficient c_n in Eq. (4), in order to study its influence on the charge motion and the power draw: $\varepsilon_n = 0.3$, 0.6 and 0.9. Since a coefficient of restitution takes a value between 0 and 1, the choice of these three values seems to be sufficient to draw a conclusion for the entire interval. Practically speaking, these values might seem to be very large but one should bear in mind that this coefficient does not depend on the impact speed in the model while it, however, depends on it in the reality. Hence, the real coefficient of restitution might be rather important for small impact velocities with a strong elastic response (like in the mill) and rather small for important impact velocities due to the energy dissipation by plastic deformation. Again, only

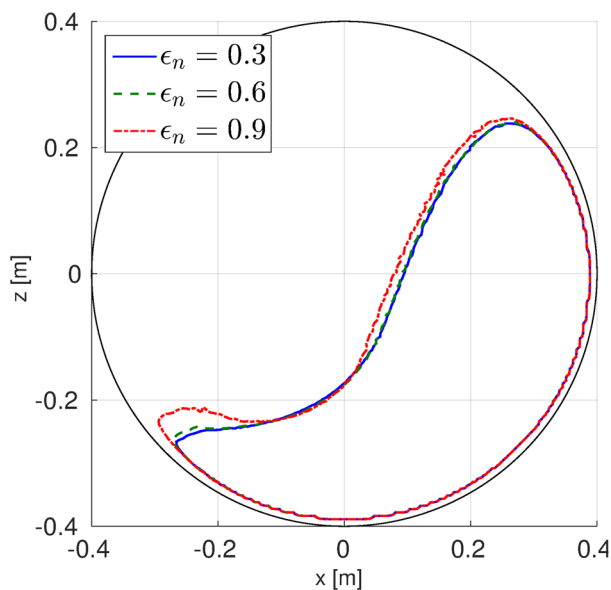


Fig. 12 Influence of the coefficient of normal restitution ϵ_n on the global charge motion. This figure shows the outline of the mill shell and the position density limits for different values of the coefficient of restitution as well as for unchanged values of the normal stiffness coefficient $k_n = 10^5$ N/m and the coefficient of friction $\mu = 0.75$. (Color figure online)

the case with the standard parameters in Table 2 is studied but identical conclusions can be obtained for the possible combinations of $k_n = 10^5$, 10^6 and 10^7 N/m as well as $\mu = 0.25$, 0.5 and 0.75.

Figure 12 shows that the influence of the coefficient of normal restitution ϵ_n becomes important for values close to 1; apart from that, the charge motion is again relatively insensitive to the contact parameters. This behavior can simply be explained by the fact that if a ball loses less energy during a collision, it is able to rebound more powerfully. Therefore, the charge cloud expands with the coefficient of restitution. This expansion is particularly important in the toe region, where the liner profile reenters into the charge and thus transfers a significant amount of energy to the balls. Since the charge is on average displaced to the left, when the coefficient of restitution increases, its center of gravity gets closer to the horizontal center of the mill. Thus, the power draw decreases with ϵ_n : the average power draw is equal to 1391, 1393 and 1379 W, for $\epsilon_n = 0.3$, 0.6 and 0.9. The second value of the power draw again does not follow the trend because of the limited sampling duration; this incoherence is, however, negligible which is the reason why this duration was not increased. Globally, the variation of the power draw with the coefficient of restitution remains relatively small (around 1 %), even though this coefficient almost takes its extremal values. In fact, losing the energy in a small number of collisions or a large number essentially has the same effect: the available energy is dissipated [3].

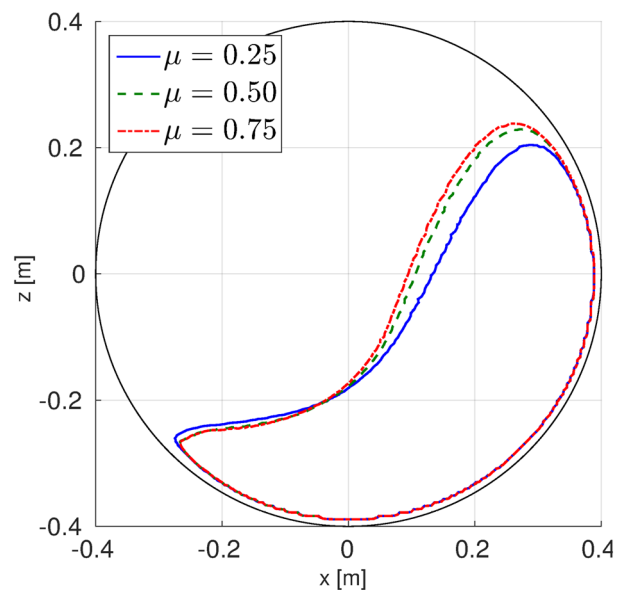


Fig. 13 Influence of the coefficient of friction μ on the global charge motion. This figure shows the outline of the mill shell and the position density limits for different values of the coefficient of friction as well as for unchanged values of the normal stiffness coefficient $k_n = 10^5$ N/m and the coefficient of normal restitution $\epsilon_n = 0.3$. (Color figure online)

3.1.4 Influence of the friction

While the stiffness and the coefficient of restitution only have a relatively small impact on the power draw, it varies strongly with the coefficient of friction μ . For instance, if $\mu = 0.25$, 0.5 and 0.75 while the other contact parameters are determined as described in Table 2, the power draw is equal to 1311, 1382 and 1391 W, respectively. The reason behind this important influence is simply the supplementary sliding and the resulting reduction of the charge lifting, when the coefficient of friction decreases. Hence, the shoulder decreases and the toe increases at the same time as shown in Fig. 13.

The shoulder region is, however, more influenced by the coefficient of friction than the toe region due to the compensation of two antagonistic effects in the toe region. On the one hand, the toe position should decrease, when the coefficient of friction increases, since the liner drags more strongly the balls away from the toe region. On the other hand, the balls fly higher in the shoulder region due to the additional momentum transfer from the liner to the balls, when the coefficient of friction increases. Hence, the balls cascade more powerfully to the toe region. The combination of these effects might explain the relatively low sensitivity of the toe region to the coefficient of friction.

3.2 Experimental calibration and validation

In this section, the contact parameters will finally be calibrated based on the experimental data of the case 1 described

in Table 1. To illustrate the predictive capability of the model, it is then validated based on the experimental data of the cases 2, 3 and 4. Unless otherwise stated, the values of the parameters are the same than those defined at the beginning of the previous section (Sect. 3.1).

3.2.1 Calibration

To calibrate the model, the numerical material parameters, which lead to the best correlation between the theoretical and experimental results, have to be determined.

First, it is inefficient and unnecessary to use unreasonably high stiffness values, like 10^6 and 10^7 N/m in this case, even though the predicted power might be slightly lower than with important stiffness values (Sect. 3.1.2). Simultaneously, choosing a moderate stiffness does not mean choosing a very small stiffness. Figure 10 showed that the overlap rises quickly to non negligible values, when the stiffness k_n drops below 10^5 N/m. Hence, $k_n = 10^5$ N/m is an acceptable limit, which minimizes the computation time.

In terms of the power draw, the laboratory mill consumes approximately 1449 W in case 1. Since the ball charge weights around 250 kg, it can be expected that around 1 to 5 % of this power are dissipated by viscous dissipation in the bearings of the mill, by damped vibrations, ..., which are obviously nonexistent in the simulation. Therefore, the predicted power draw should have a value between 1376 and 1434 W in this case. In consequence, it is necessary that the coefficient of friction is at least equal to roughly 0.75, which is a realistic value. Otherwise, such high values of power draw could not be obtained.

In order to select a coefficient of restitution with $k_n = 10^5$ N/m and $\mu = 0.75$, the position density limits for different coefficients of restitution were superimposed on a photograph of the charge in Fig. 14. Due to the pronounced perspective of the picture and the parallel projection of the charge in the position density plot, it is necessary that the contour only delimits the charge in contact with the transparent end walls, even though this charge is obviously lifted a little higher than the charge at the center of the mill due to the contact with the end wall.

For all four coefficients of normal restitution, Fig. 14 shows a very good correlation between the experimental and the simulated charge motions. For $\epsilon_n = 0.3$ and 0.6, the toe angle seems, however, to be slightly too small, while it is too important for $\epsilon_n = 0.9$. Hence, the contour curve for $\epsilon_n = 0.8$ was added to find a better compromise. The same value was used by Cleary to simulate the balls in a cement tube mill without feed material [6]. This choice is also more reasonable than $\epsilon_n = 0.3$ (used in previous studies in the presence of the feed material), for balls and liner plates made from steel, which probably remain fairly elastic and thus energy-preserving during a collision. The predicted power

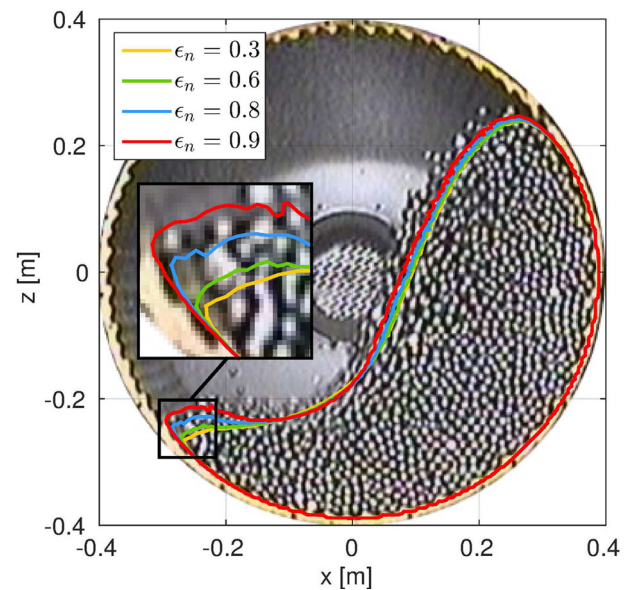


Fig. 14 Ball charge photograph with the superimposed position density limit curves for different coefficients of normal restitution ϵ_n . The normal stiffness k_n and the coefficient of friction μ are equal to 10^5 N/m and 0.75, respectively. The other parameter values can be deduced from these values thanks to Table 2. The position density limits might not be clearly visible if the paper was printed in black-and-white. (Color figure online)

draw is equal to 1394 W, in this case, i.e. for $k_n = 10^5$ N/m, $\epsilon_n = 0.8$ and $\mu = 0.75$. Thus, the predicted power is 3.8 % lower than the real power draw, which might be explained by the non-existence of some specific dissipation mechanisms in the model, as explained earlier in Sect. 3.2.1. Since determining this power loss accurately is quite complex and since a 3.8 %-error remains negligible with respect to the stochastic fluctuations of the power draw in the quasi-steady state, this last set of material parameters is as close as one can get to the precise prediction of the charge motion and the power draw of the laboratory mill.

3.2.2 Validation

To prove that the model is truly predictive with the material parameter values of the previous section ($k_n = 10^5$ N/m, $\epsilon_n = 0.8$, $\mu = 0.75$) and to determine whether the position density limit describes accurately the global charge motion, the present model will be used to estimate the charge motion and the power draw of the different experimental scenarios described in Table 1. The results, i.e. the position density plots, the position density limits, the superposed charge motion photographs and the power draw comparisons are shown in Fig. 15 and in Table 3.

Essentially, two conclusions can be drawn from this figure and the table. The most important conclusion is certainly that the DEM model is truly *predictive* with respect to the global charge motion. In fact, it is difficult to see any systematic

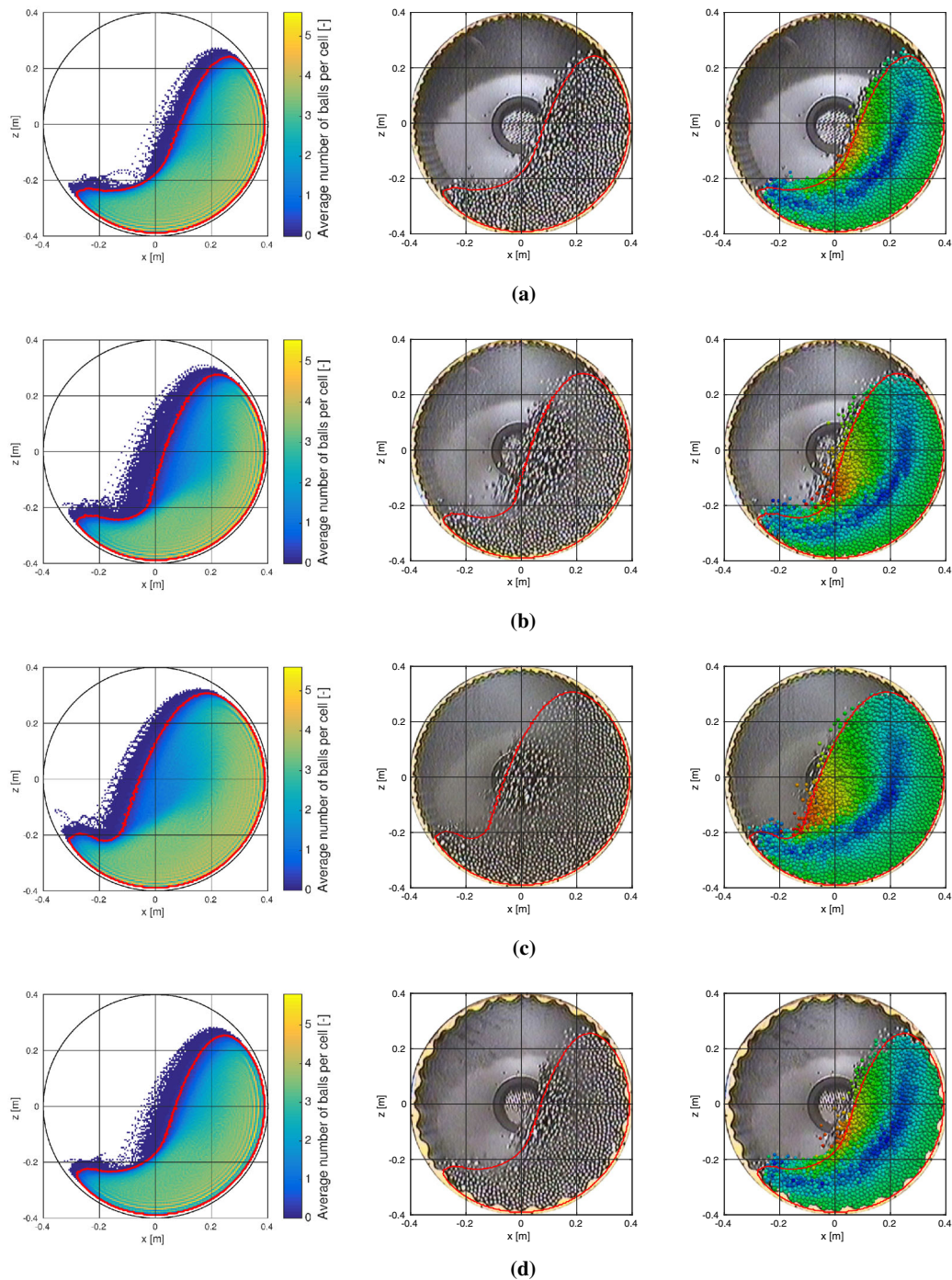


Fig. 15 Validation of the ball mill DEM model. *Left* position density plot and position density limit to show that the choice of the specific contour level was well chosen. *Middle* position of the real charge and the position density limit of the simulation for the respective case. *Right* position of the real charge and the superimposed simulated charge after 5 s of simulated time. The color code represents the amplitude of the translation velocity of each ball, i.e. *blue* (at rest) to *red* (about 3 m/s). The velocity as such has no real importance here except for the different

color, which renders it possible to distinguish the information from the two superimposed pictures. The position density limits and the superimposed ball charge might not be clearly visible if the paper was printed in black-and-white. **a** Case 1: liner 1, 30 % filling ratio, 70 % of critical velocity. **b** Case 2: liner 1, 30 % filling ratio, 80 % of critical velocity. **c** Case 3: liner 1, 40 % filling ratio, 80 % of critical velocity. **d** Case 4: liner 2, 30 % filling ratio, 70 % of critical velocity. (Color figure online)

Table 3 Average real power draw P_{real} and average predicted power draw P_{pred} of the laboratory mill in the quasi-steady state for the different calibration and validation cases from Table 1. The relative error E_r with respect to the real power draw was also calculated

Case	P_{real} (W)	P_{pred} (W)	E_r (%)
Case 1	1449	1394	3.8
Case 2	1656	1630	1.6
Case 3	1893	1864	1.5
Case 4	1429	1365	4.5

difference between the real ball charge, the position density limit and the simulated ball charge. The second conclusion is that the *position density limit* is a perfect objective tool to describe the global charge motion and in particular, to determine the shoulder and toe angles based on the DEM model.

Concerning the average power draw, Table 3 shows that the real power draw is very close to the predicted power draw. More precisely, the real power draw is systematically greater than the predicted power draw, which might be explained by the additional dissipation mechanisms in the reality. The relative error between the predicted and the real power draw is, however, always smaller than 5 %, which is negligible in practice.

Hence, the DEM model is able to predict the global charge motion and the power draw of the laboratory mill for different filling ratios, different percentages of critical velocity and different liner profiles within the limit of their validity (photographs of the charge motion and variation of the power draw by 5 %). Therefore, the present DEM model renders future experimental testing unnecessary within these limits.

4 Conclusion

Discrete element method simulations of a 1:5-scale laboratory ball mill have been performed to study the influence of the contact parameters on the charge motion and the power draw of the mill.

The experimental setup consisted of a 0.8 m diameter laboratory mill. This mill was filled with several thousand spherical metal balls and it was rotating at about 70 % of its critical speed in order to reproduce industrial operating conditions. Photographs of the charge motion were taken through one end wall of the mill, which was made of tempered glass. Its power draw in the quasi-steady state was measured by means of a torquemeter. The charge motion and the power draw was also determined by the 3D dis-

crete element method implemented in the software YADE. The interaction between discrete bodies was modeled by the linear spring-slider-damper contact law. The position density limit was introduced as a mathematical tool enabling a clear and objective description of the macroscopic charge motion in different scenarios; it is a specific contour curve of the position density plot, which separates the main charge zone from the region where the balls are less likely to occur.

Based on the position density limit and the power draw prediction by the explicit torque calculation in the software, it was possible to study the influence of the contact parameters on these characteristics. First, it was shown that the tangential spring and damper have no significant influence on the global charge motion and the power draw. Due to the limited calibration data, they were then assumed to be equal to their normal counterpart in the DEM simulation. Likewise, the charge motion and the power draw are also relatively insensitive to the normal stiffness provided that the maximum overlap percentage of two colliding spheres is smaller than about 10 %. The influence of the coefficient of normal restitution on the charge motion becomes important for a relatively high restitution because a ball, which loses less energy in a collision, rebounds more powerfully. The variation of the power draw with the coefficient of restitution remains, however, relatively small since losing the energy in a small number of collisions or a large number fundamentally has the same effect. Finally, the coefficient of friction has a strong influence on the charge motion and the power draw since it controls directly the sliding propensity of the charge. Hence, the shoulder position and the power draw increase with the coefficient of friction.

The previous arguments rendered it possible to calibrate the contact parameters in the simulation of the laboratory mill. Some validation cases then showed that the position density limit is an accurate tool to describe the global charge motion and that the DEM model is able to predict this charge motion and the power draw of the laboratory mill for different filling ratios, different percentages of critical velocity and different liner profiles. The discrete element method thus rendered future experimental testing in the laboratory mill unnecessary.

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Conflict of interest The authors declare that they have no conflict of interest.

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